In [1]:
```
from sklearn.datasets import load_iris
from sklearn.pipeline import Pipeline
from sklearn.model_selection import (train_test_split,cross_val_score,Gri
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score,classification_report,confusio
```

In [2]:
```
iris = load_iris()
X = iris.data
Y= iris.target
```

In [4]:
```
X_train,X_test,Y_train,Y_test = train_test_split(X,Y, test_size = 0.2, ra
```

In [5]:
```
pipeline = Pipeline([
    ("scaler",StandardScaler()),
    ("classifier",LogisticRegression(max_iter=1000))
    ])
```

In [6]:
```
cv_scores = cross_val_score(pipeline,X_train,Y_train,cv=5,scoring="accura

print("Cross-Validation Scores:",cv_scores)
print("Mean CV Accuracy:",round(cv_scores.mean(),4))
```

```
Cross-Validation Scores: [0.91666667 0.95833333 0.95833333 0.95833333 1.
]
Mean CV Accuracy: 0.9583
```

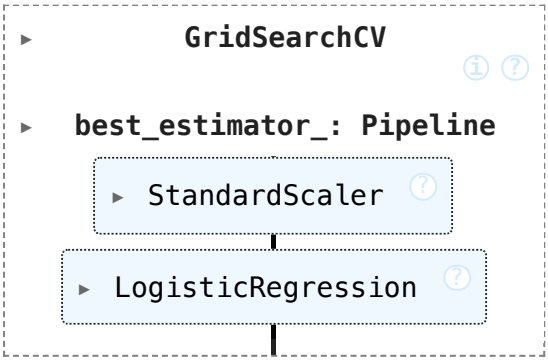
In [7]:
```
pipeline.fit(X_train,Y_train)
baseline_pred=pipeline.predict(X_test)

baseline_accuarcy= accuracy_score(Y_test,baseline_pred)
print("Baseline Test accuarcy:", round(baseline_accuarcy*100,2),"%")
```

```
Baseline Test accuarcy: 93.33 %
```

In [8]:
```
param_grid = {"classifier__C": [0.01,0.1,1,10,100]}
grid_search = GridSearchCV(
    estimator=pipeline,
    param_grid=param_grid,
    cv=5,
    scoring="accuracy",
    n_jobs=-1
)
grid_search.fit(X_train,Y_train)
```

Out[8]:



In [9]:
```
best_model=grid_search.best_estimator_

print("\nBest Parameters:",grid_search.best_params_)
print("Best CV Accuarcy:",round(grid_search.best_score_,4))
```

```
Best Parameters: {'classifier__C': 10}
Best CV Accuarcy: 0.9667
```

In [10]:
```
optimized_pred=best_model.predict(X_test)

optimized_accuracy=accuracy_score(Y_test, optimized_pred)

print("\nOptimized Test Accuarcy:",round(optimized_accuracy*100,2),"%")
```

```
Optimized Test Accuarcy: 100.0 %
```

In [11]:
```
print("\nBaseline  accuarcy:", round(baseline_accuarcy*100,2),"%")
print("\nOptimized  Accuarcy:",round(optimized_accuracy*100,2),"%")
```

```
Baseline  accuarcy: 93.33 %

Optimized  Accuarcy: 100.0 %
```

In [ ]: